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ManiSkill3: GPU-Parallelized Robotics Simulation and Rendering for Generalizable Embodied AI

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Abstract

Simulation has enabled unprecedented compute-scalable approaches to robot learning. ManiSkill3 is the fastest state-visual GPU-parallelized robotics simulator with contact-rich physics, supporting heterogeneous simulation of diverse manipulation tasks, scenes, and robot embodiments for generalizable embodied AI. ManiSkill3 trains manipulation policies 10-1000x faster than alternatives while preserving physical accuracy and supporting realistic ray-tracing for sim-to-real transfer. The benchmark covers over 20 task families, including dexterous in-hand manipulation, mobile manipulation, locomotion, and humanoid whole-body control.

Introduction

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Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2410.00425, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for ManiSkill3: GPU-Parallelized Robotics Simulation and Rendering for Generalizable Embodied AI follows the abstract: Simulation has enabled unprecedented compute-scalable approaches to robot learning. ManiSkill3 is the fastest state-visual GPU-parallelized robotics simulator with contact-rich physics, supporting heterogeneous simulation of diverse manipulation tasks, scenes, and robot embodiments for generalizable embodied AI. ManiSkill3 trains manipulation policies 10-1000x faster than alternatives while preserving physical accuracy and supporting realistic ray-tracing for sim-to-real transfer. The benchmark covers over 20 task families, including dexterous in-hand manipulation, mobile manipulation, locomotion, and humanoid whole-body control. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: Simulation has enabled unprecedented compute-scalable approaches to robot learning. ManiSkill3 is the fastest state-visual GPU-parallelized robotics simulator with contact-rich physics, supporting heterogeneous simulation of diverse manipulation tasks, scenes, and robot embodiments for generalizable embodied AI. ManiSkill3 trains manipulation policies 10-1000x faster than alternatives while preserving physical accuracy and supporting realistic ray-tracing for sim-to-real transfer. The benchmark covers over 20 task families, including dexterous in-hand manipulation, mobile manipulation, locomotion, and humanoid whole-body control.

Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

Conclusion

We conclude by summarizing the contribution of arXiv:2410.00425 and outline directions for follow-up work.

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References

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